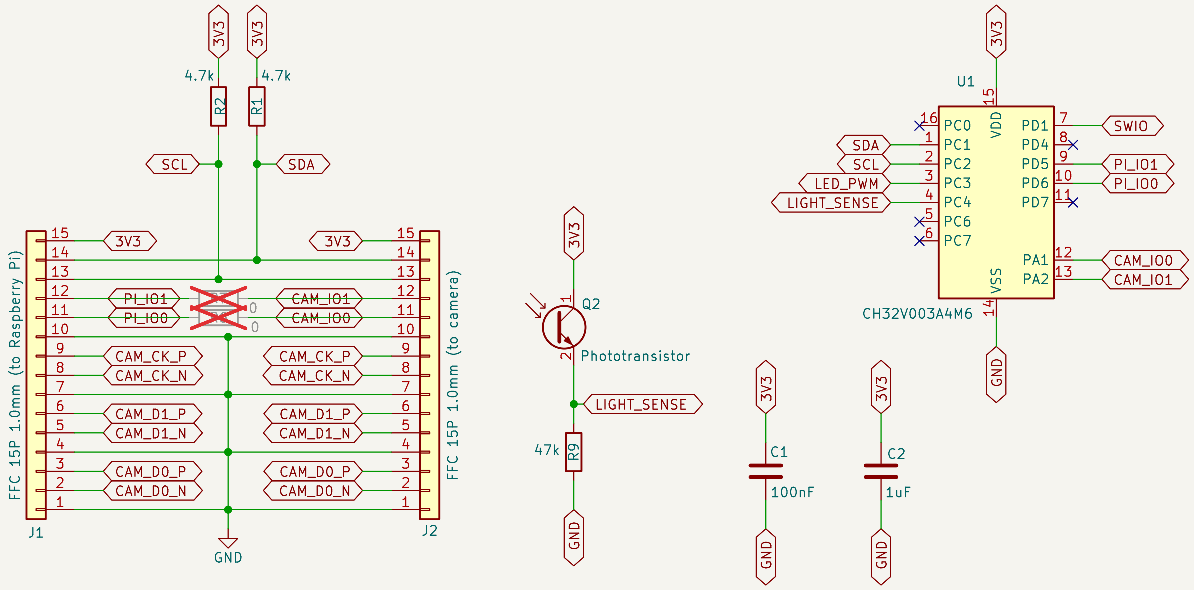
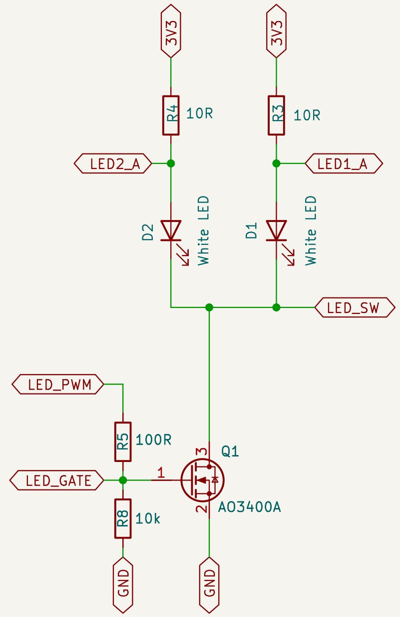
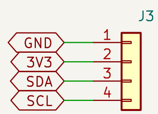
RPi Camera LED interposer: sits between a Raspberry Pi and its camera.
MIPI lanes, I2C, GND and 3V3 pass straight through J1 -> J2.
U1 (I2C slave on the camera bus) drives the camera GPIO pins 11/12,
PWM-drives the illumination LEDs and reads ambient light on its ADC.



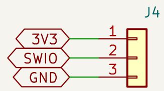
LED current is set by R3/R4 against the 3V3 rail (about 30-40mA each
with Vf=2.9V). Total draw comes from the Pi camera 3V3 supply - keep
the duty/current within the Pi's camera connector budget.



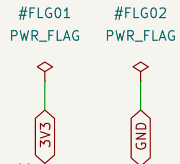
R6/R7 (0R, DNP) are a bypass: fit them to wire the Pi's CAM_IO pins
straight to the camera, e.g. when U1 is unpopulated. Left off, U1 sits in
the middle -- it watches the Pi on PD5/PD6 and drives the camera on PA1/PA2.
Do not fit them while U1 drives: 0R gives contention no current limit.



Qwiic (JST SH 1.0mm)



SWD program header 2.54mm



All parts chosen for JLCPCB / NextPCB Rev0 assembly; see
research/*.md for availability and pricing (checked 2026-09-01).

SOP-16 bonds no two ports to one pin, so PD1/SWIO is programming only
and PC3/T1CH3 drives the LED gate: no LED flicker while flashing firmware.
PC6, PC7, PD4 and PC0 are spare. PD7 is left unused on purpose: it is NRST
until the RST_MODE option byte says otherwise, so a Pi GPIO on it would
reset the MCU on a factory-fresh part.

CH32V003A4M6 camera GPIO / illumination / light sensor

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Title: RPi Camera LED interposer

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